US 20200324794A1

(19) **United States**

(12) **Patent Application Publication**

(10) **Pub. No.: US 2020/0324794 A1**

(43) **Pub. Date: Oct. 15, 2020**

**Ma et al.**

(54) **TECHNOLOGY TO APPLY DRIVING NORMS FOR AUTOMATED VEHICLE BEHAVIOR PREDICTION**

(71) Applicant: **Intel Corporation**, Santa Clara, CA (US)

(72) Inventors: **Guixiang Ma**, Hillsboro, OR (US); **Nicole Beckage**, Beaverton, OR (US); **Nesreen Ahmed**, Santa Clara, CA (US); **Ignacio Alvarez**, Portland, OR (US)

(21) Appl. No.: **16/912,241**

(22) Filed: **Jun. 25, 2020**

**Publication Classification**

(51) **Int. Cl.**

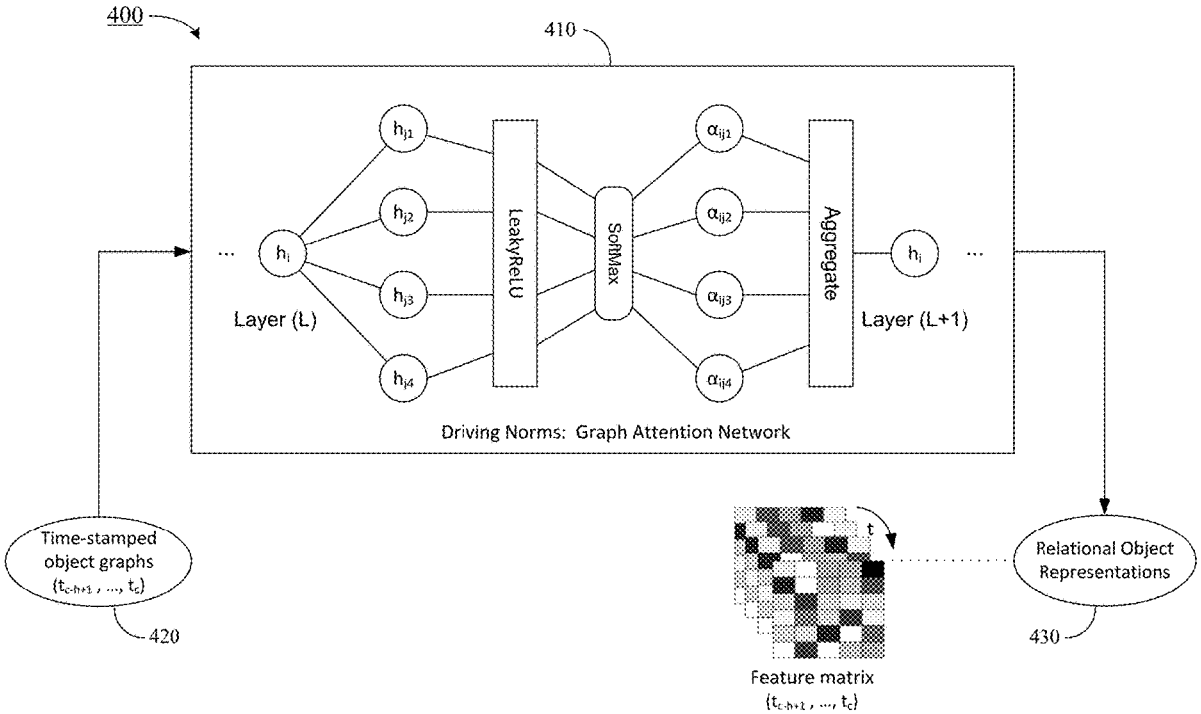
| | |
|---|---|
| ***B60W 60/00*** | (2006.01) |
| ***G06N 3/04*** | (2006.01) |
| ***G06N 3/08*** | (2006.01) |
| ***G06F 16/28*** | (2006.01) |
| ***G06F 16/29*** | (2006.01) |

(52) **U.S. Cl.**

CPC .. ***B60W 60/00272*** (2020.02); ***B60W 60/0011*** (2020.02); ***G06N 3/0454*** (2013.01); *B60W 2050/0089* (2013.01); ***G06F 16/284*** (2019.01); ***G06F 16/29*** (2019.01); ***G06N 3/08*** (2013.01)

(57) **ABSTRACT**

Systems, apparatuses and methods may provide for technology that generates a series of time-stamped object graphs based on object trajectory histories derived from external object data for a plurality of external objects, such as vehicles. The technology may also generate, via a first neural network such as a graph attention network, a series of relational object representations based on the series of time-stamped object graphs, and determine, via a second neural network such as a long short-term memory network, predicted object trajectories for the plurality of external objects based on the series of relational object representations. The technology may also modify behavior of an autonomous vehicle based on the predicted object trajectories and real-time perceptual error information.



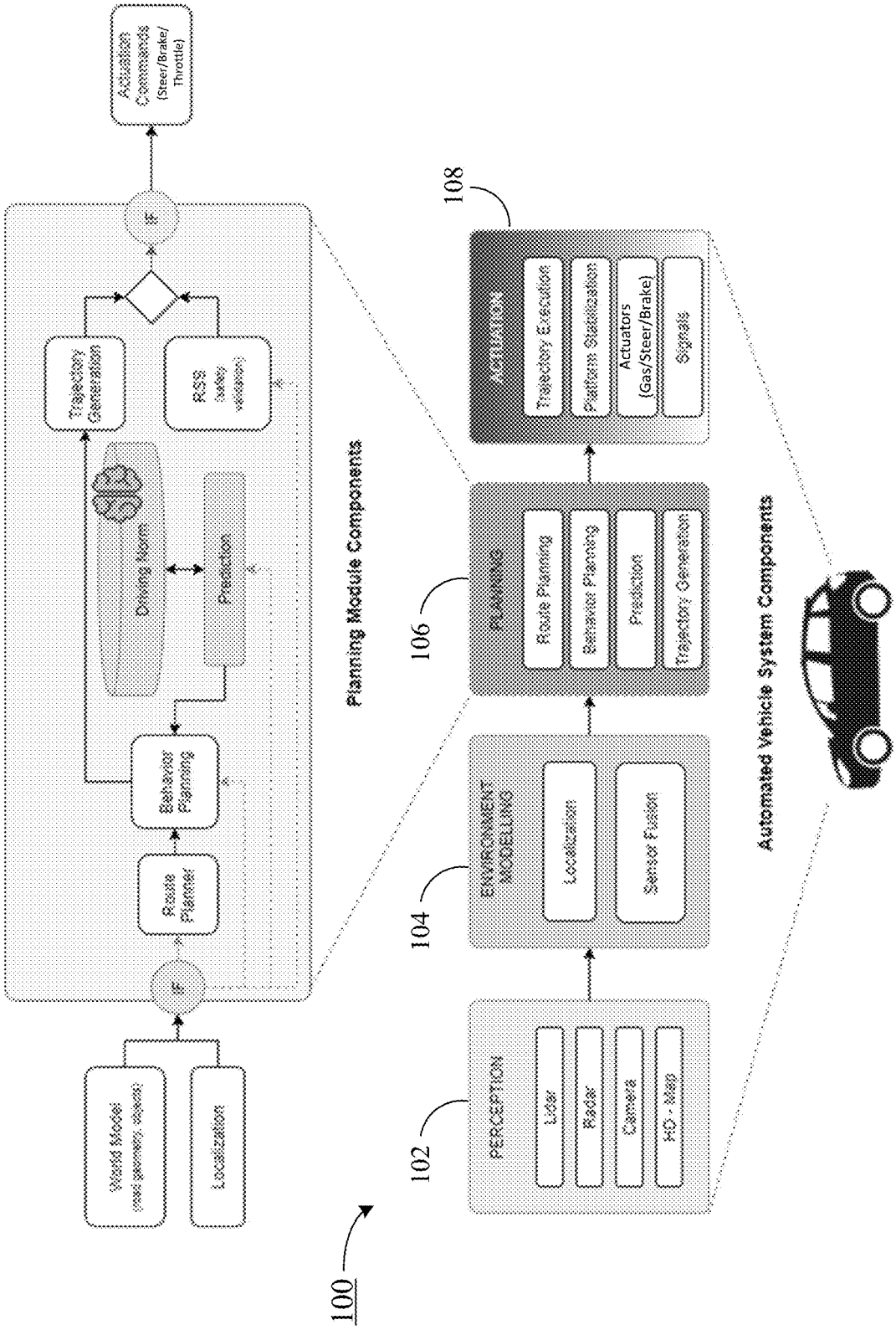

100
102
PERCEPTION
Lidar
Radar
Camera
HD - Map
104
ENVIRONMENT MODELING
Localization
Sensor Fusion
106
PLANNING
Route Planning
Behavior Planning
Prediction
Trajectory Generation
108
ACTUATION
Trajectory Execution
Platform Stabilization
Actuators (Gas/Steer/Brake)
Signals
Automated Vehicle System Components
Planning Module Components
World Model
Localization
Route Planner
Behavior Planning
Driving Norm
Prediction
Trajectory Generation
RSS
Actuation Commands (Steer/Brake/ Throttle)

FIG. 1

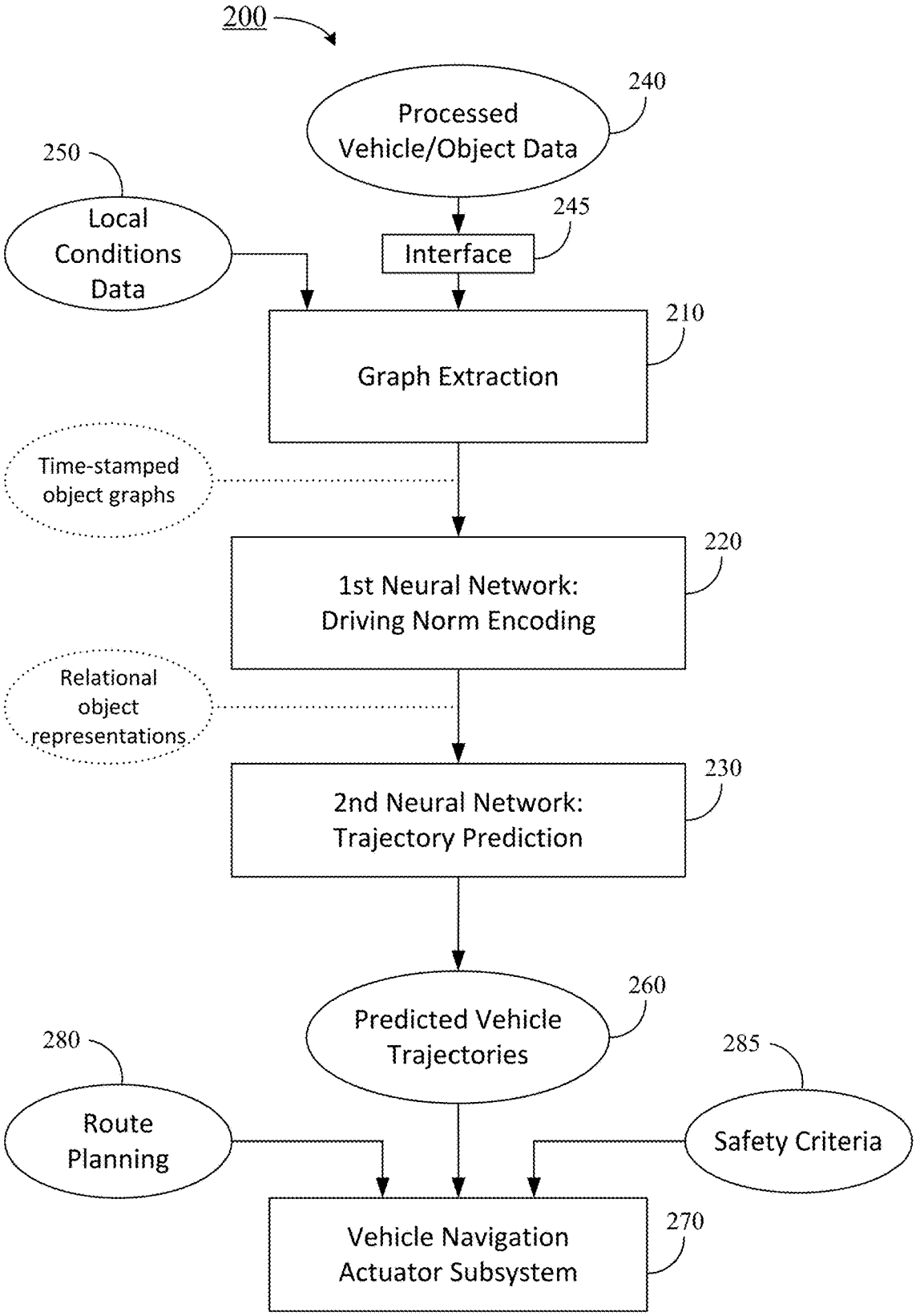

200
Processed Vehicle/Object Data
240
250
Local Conditions Data
245
Interface
210
Graph Extraction
Time-stamped object graphs
220
1st Neural Network: Driving Norm Encoding
Relational object representations
230
2nd Neural Network: Trajectory Prediction
260
Predicted Vehicle Trajectories
280
285
Route Planning
Safety Criteria
270
Vehicle Navigation Actuator Subsystem

FIG. 2

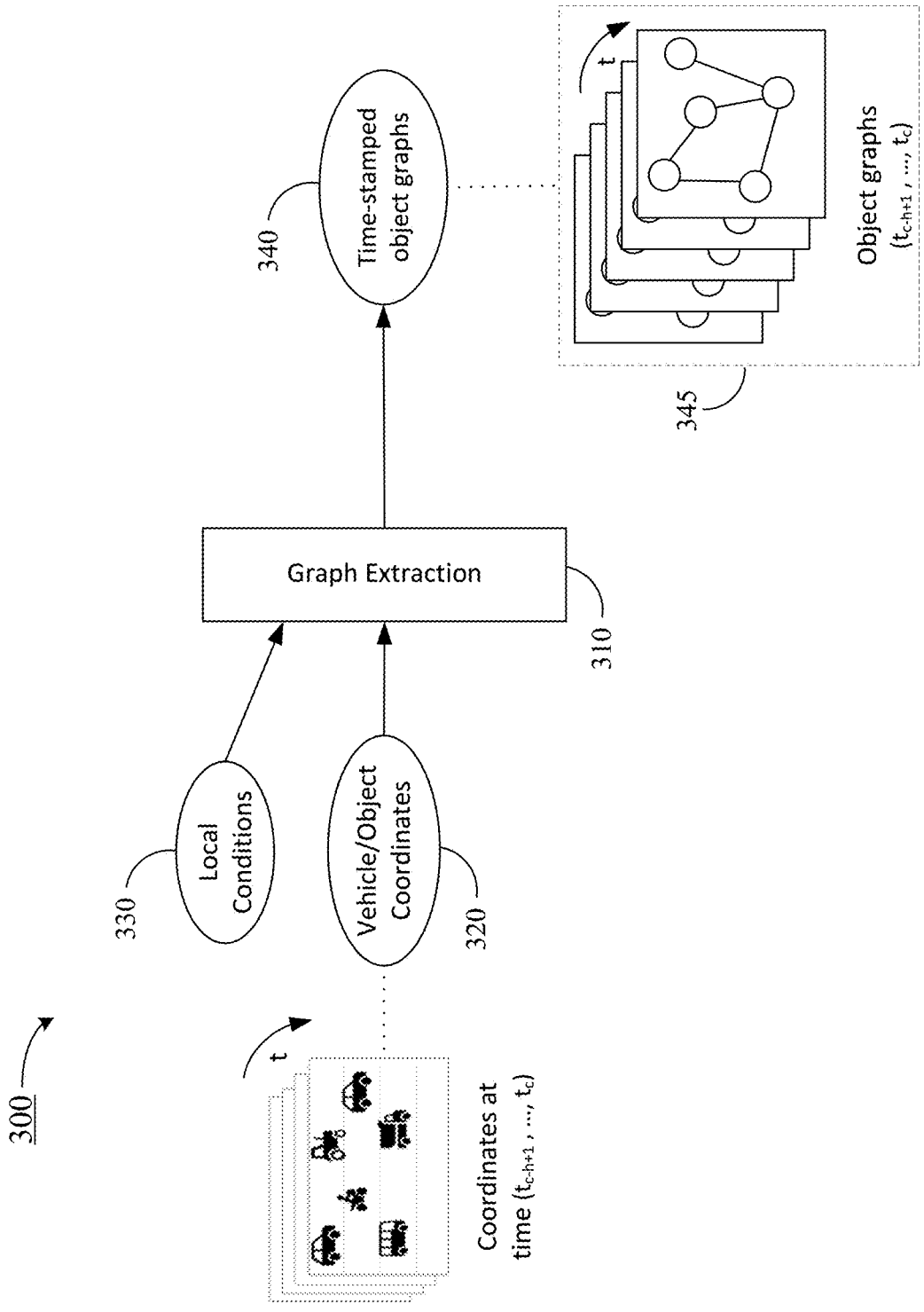

300
t
Coordinates at time ($t_{c-h+1}$, ..., $t_c$)
330
Local Conditions
320
Vehicle/Object Coordinates
Graph Extraction
310
340
Time-stamped object graphs
t
Object graphs ($t_{c-h+1}$, ..., $t_c$)
345

FIG. 3

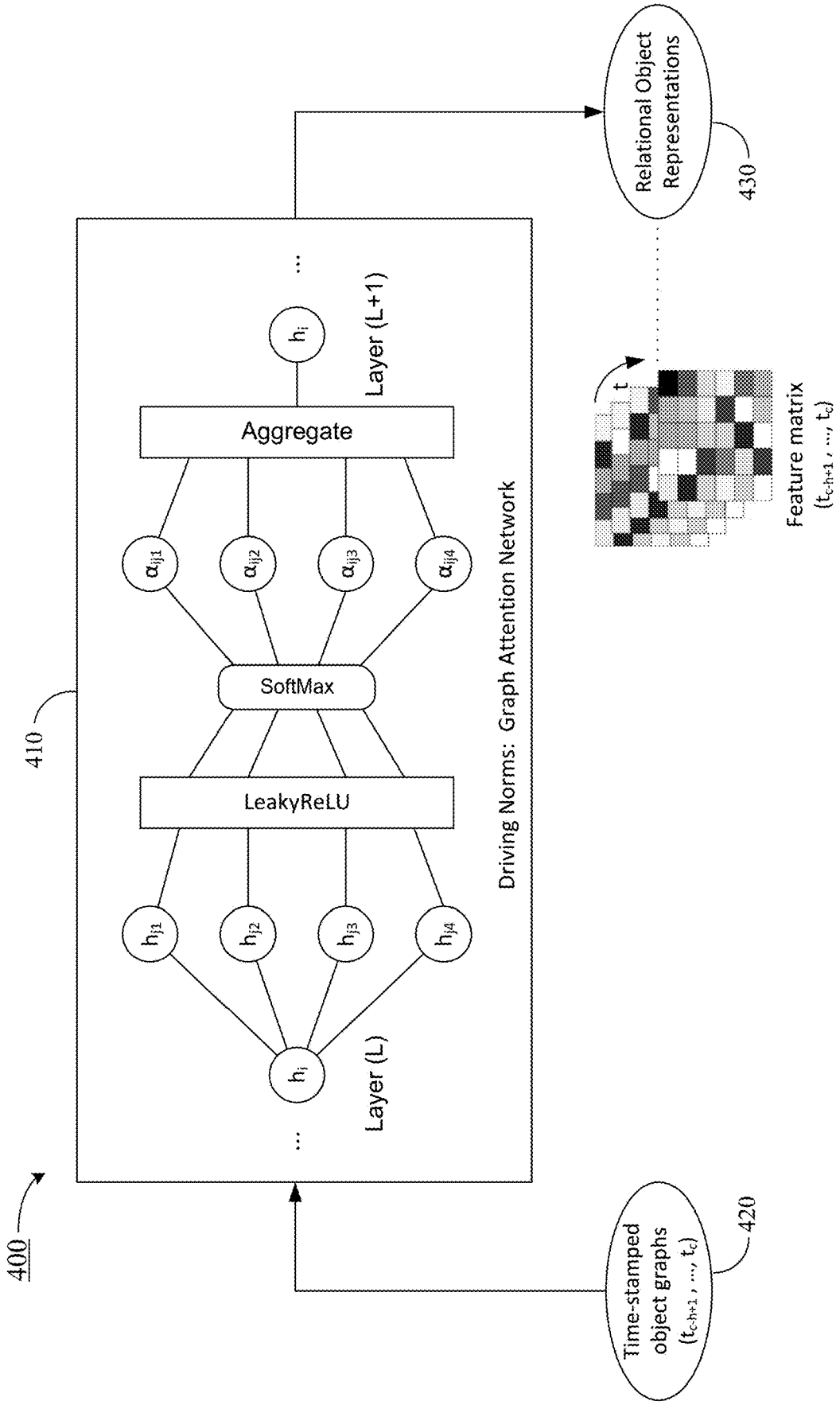

400
410
Time-stamped object graphs ($t_{c-h+1}$, ..., $t_c$)
420
Driving Norms: Graph Attention Network
Layer (L)
$h_{j1}$
$h_{j2}$
$h_{j3}$
$h_{j4}$
$h_i$
LeakyReLU
SoftMax
$\alpha_{ij1}$
$\alpha_{ij2}$
$\alpha_{ij3}$
$\alpha_{ij4}$
Aggregate
Layer (L+1)
$h_i$
Feature matrix ($t_{c-h+1}$, ..., $t_c$)
t
Relational Object Representations
430

FIG. 4

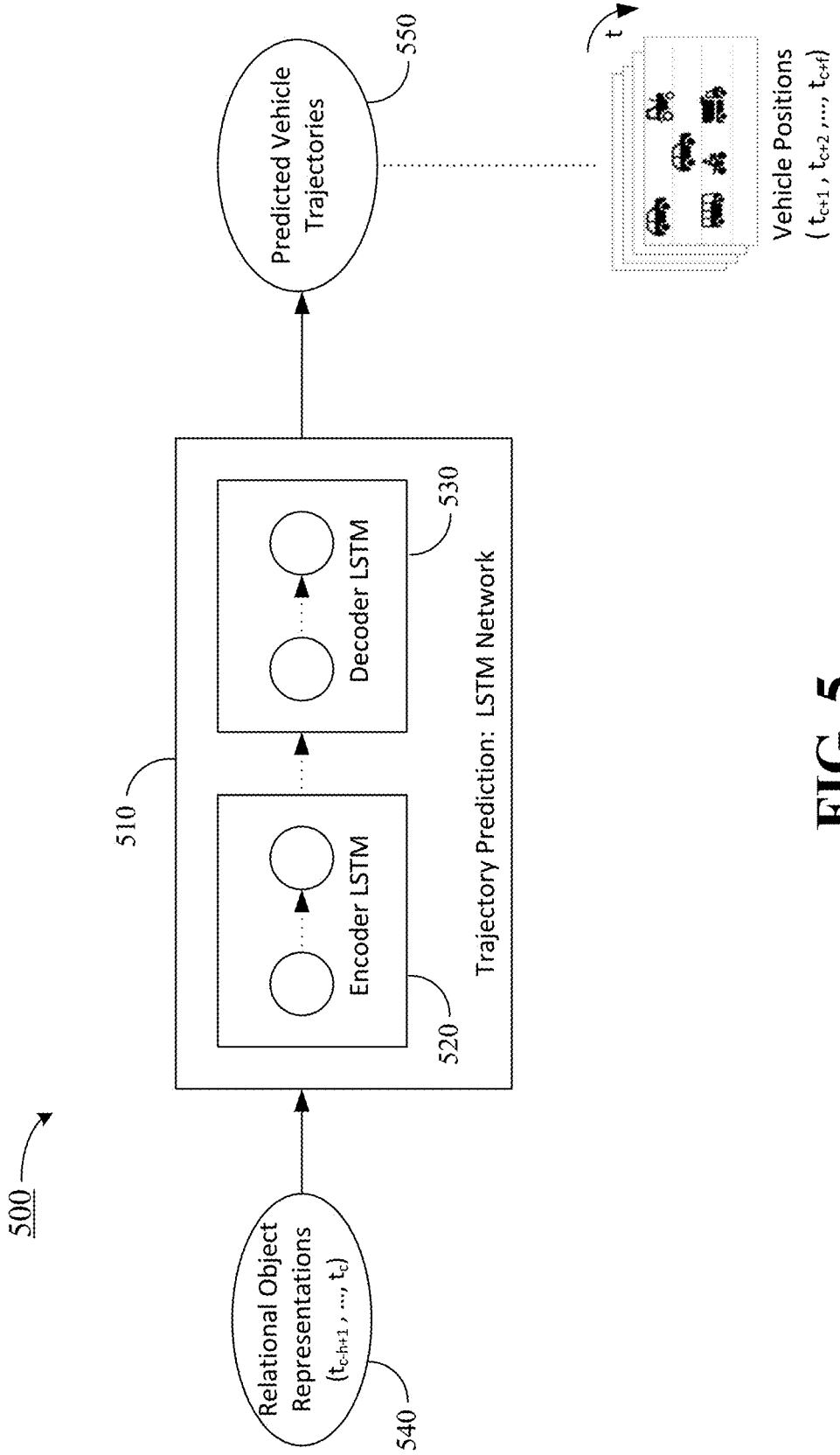

500
Relational Object Representations ($t_{c-h+1}$, ..., $t_c$)
540
Trajectory Prediction: LSTM Network
510
Encoder LSTM
520
Decoder LSTM
530
Predicted Vehicle Trajectories
550
t
Vehicle Positions ($t_{c+1}$, $t_{c+2}$, ..., $t_{c+f}$)

FIG. 5

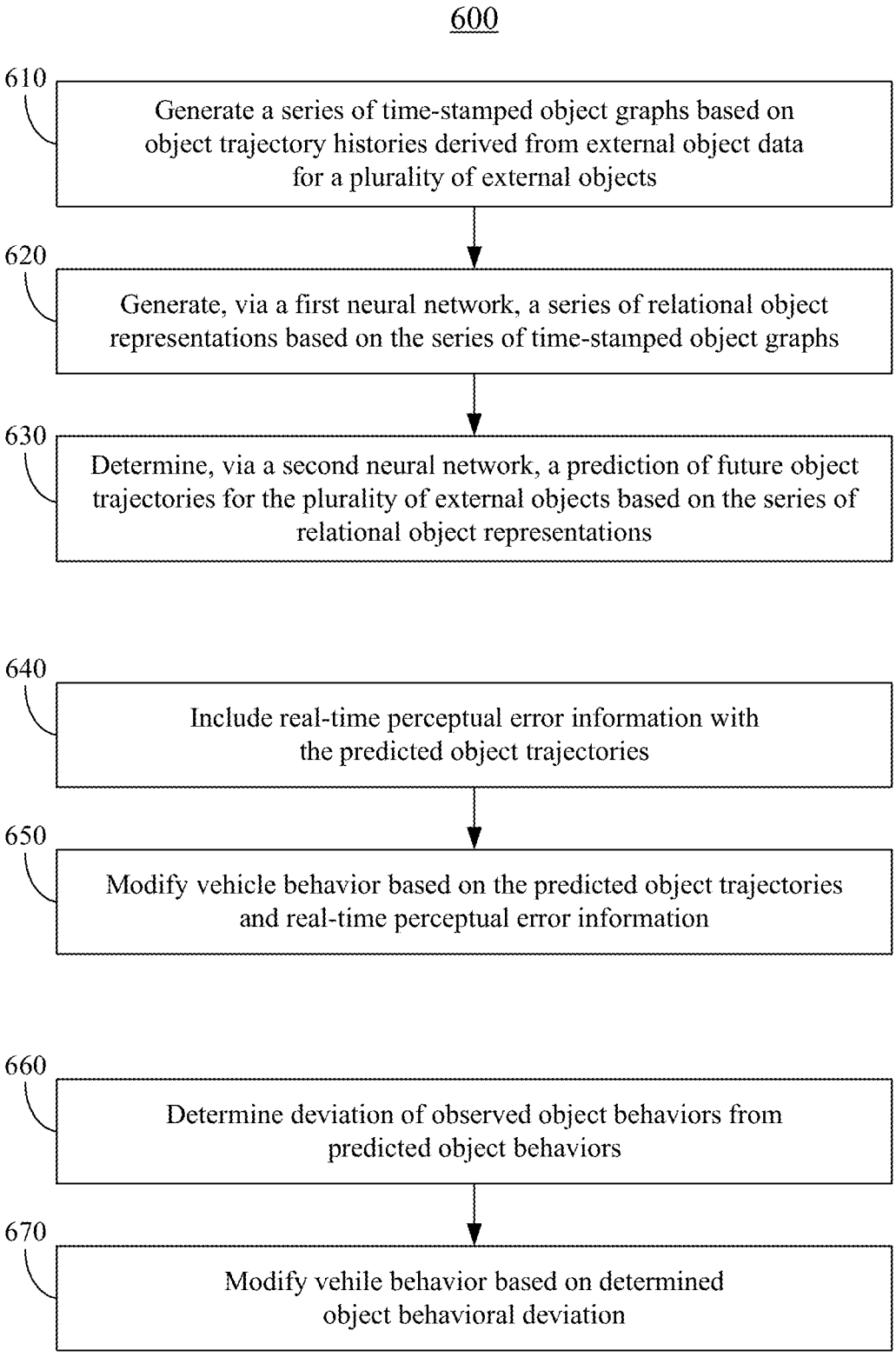

600
610
Generate a series of time-stamped object graphs based on object trajectory histories derived from external object data for a plurality of external objects
620
Generate, via a first neural network, a series of relational object representations based on the series of time-stamped object graphs
630
Determine, via a second neural network, a prediction of future object trajectories for the plurality of external objects based on the series of relational object representations
640
Include real-time perceptual error information with the predicted object trajectories
650
Modify vehicle behavior based on the predicted object trajectories and real-time perceptual error information
660
Determine deviation of observed object behaviors from predicted object behaviors
670
Modify vehile behavior based on determined object behavioral deviation

FIG. 6

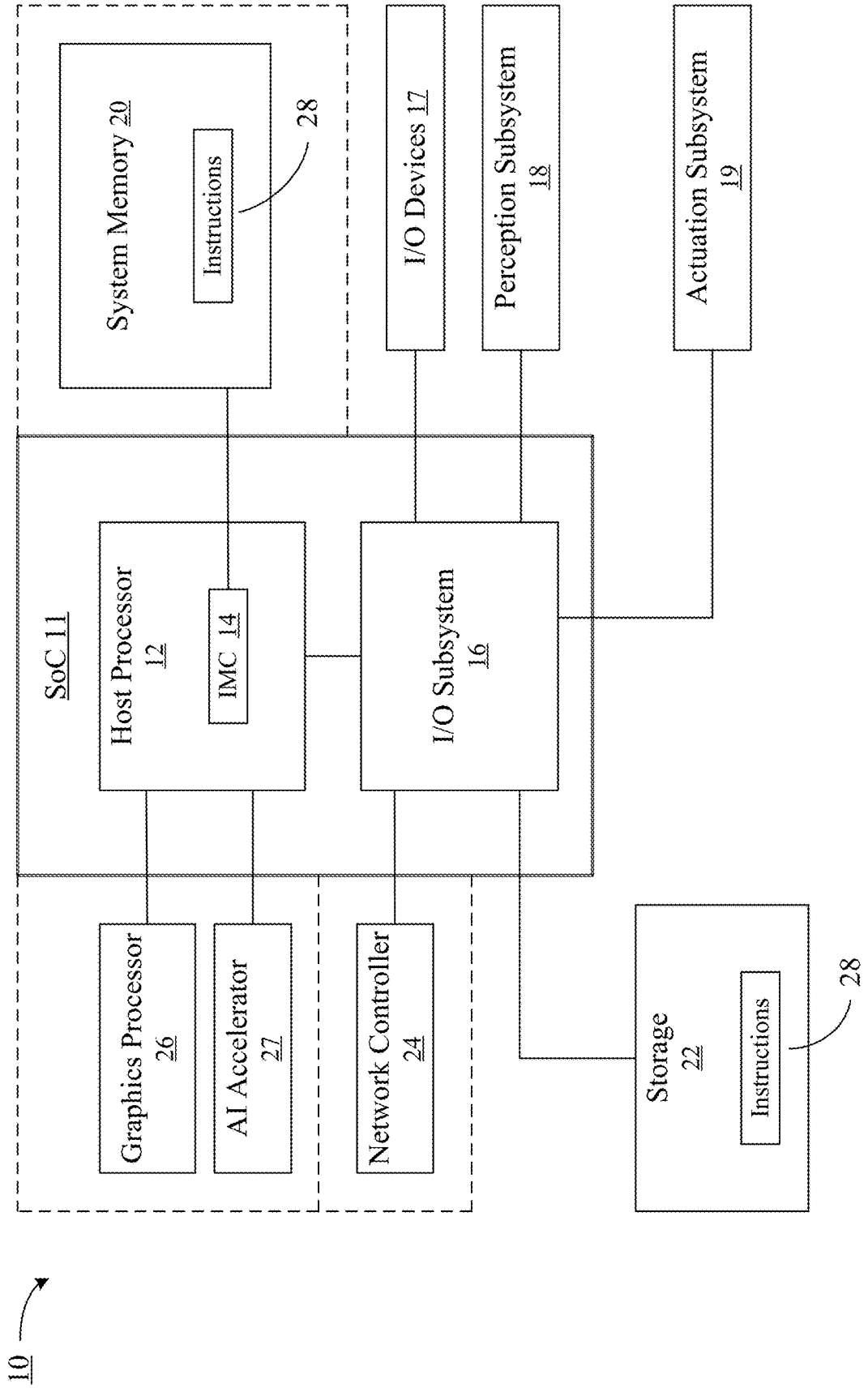

10
SoC 11
System Memory 20
Instructions
28
Host Processor 12
IMC 14
I/O Subsystem 16
I/O Devices 17
Perception Subsystem 18
Actuation Subsystem 19
Graphics Processor 26
AI Accelerator 27
Network Controller 24
Storage 22
Instructions
28

FIG. 7

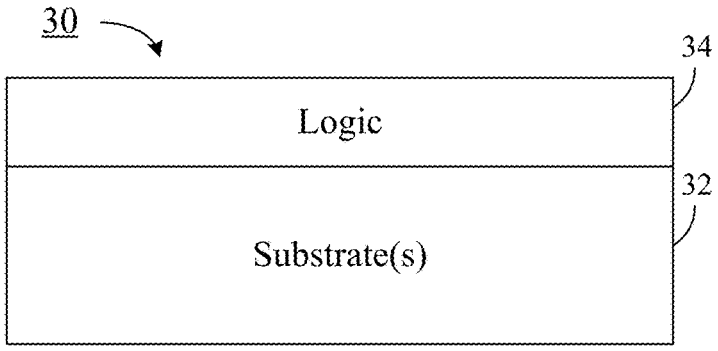

30
34
Logic
32
Substrate(s)

FIG. 8

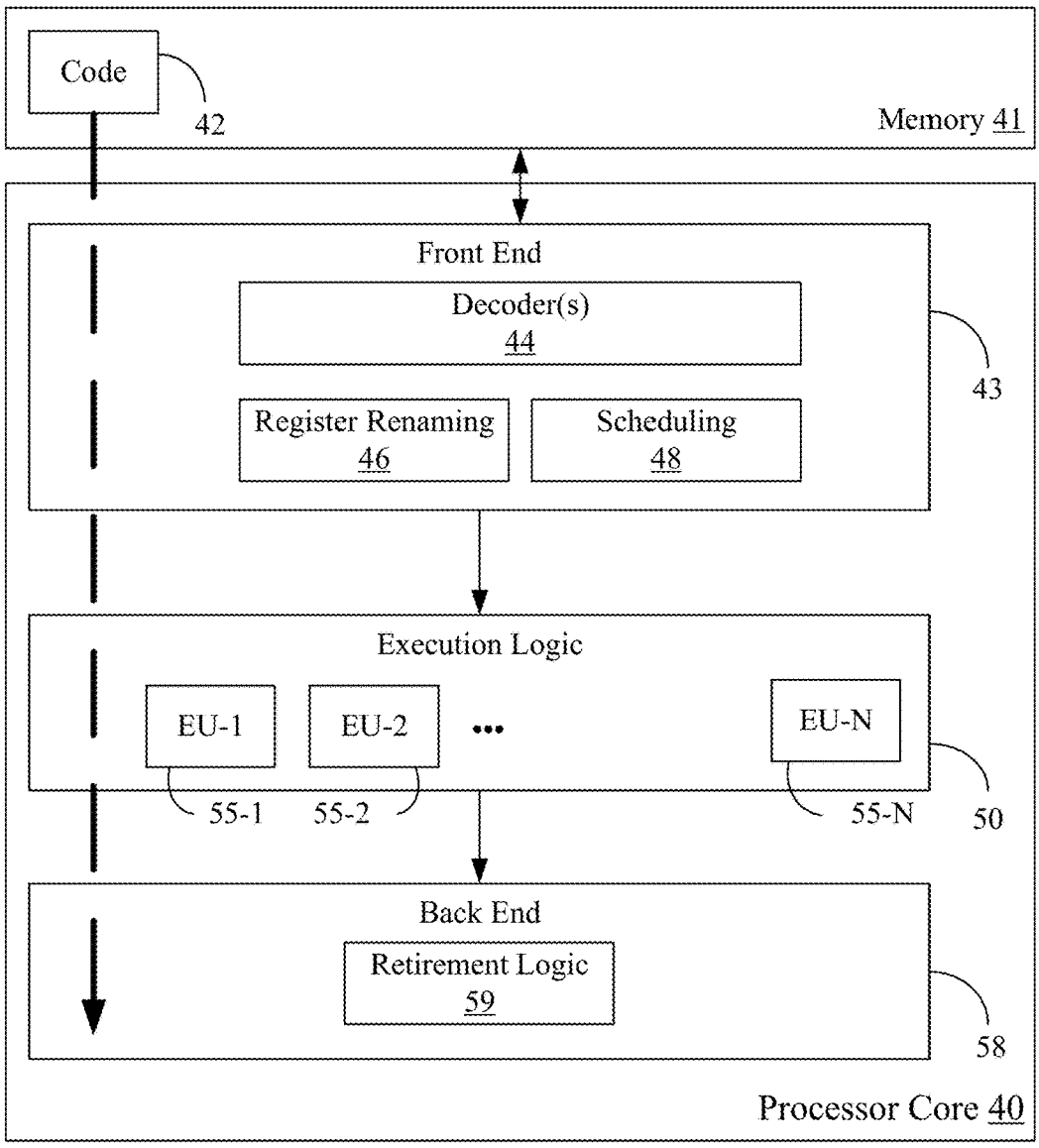
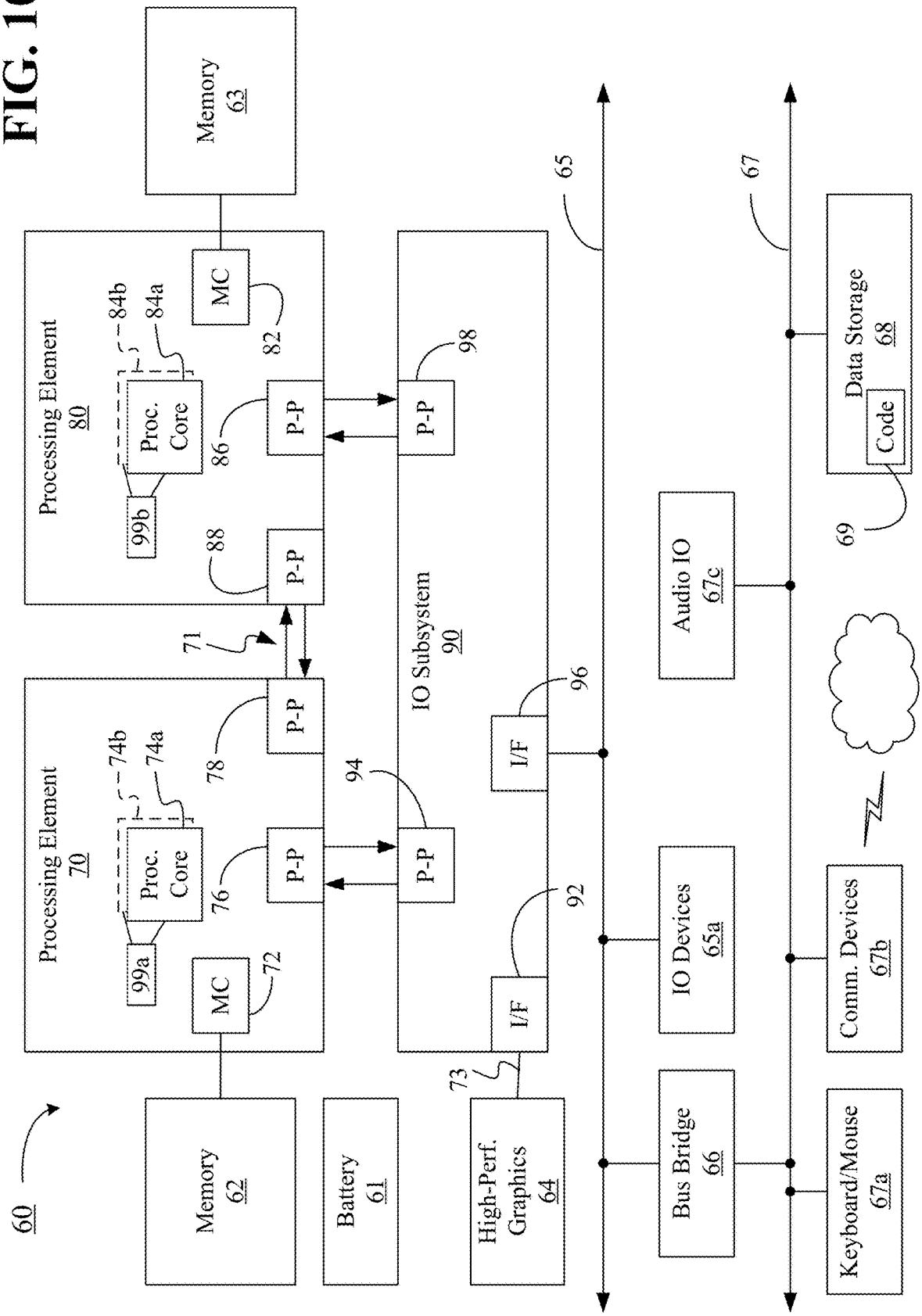

Code
42
Memory 41
Front End
Decoder(s)
44
43
Register Renaming
46
Scheduling
48
Execution Logic
EU-1
EU-2
...
EU-N
55-1
55-2
55-N
50
Back End
Retirement Logic
59
58
Processor Core 40

FIG. 9

FIG. 10


60
Memory 62
Battery 61
High-Perf. Graphics 64
Bus Bridge 66
Keyboard/Mouse 67a
IO Devices 65a
Comm. Devices 67b
Audio IO 67c
Data Storage 68
Code
69
65
67
Processing Element 70
Proc. Core
74a
74b
99a
MC
72
P-P
76
78
94
I/F
73
92
96
IO Subsystem 90
P-P
98
Processing Element 80
Proc. Core
84a
84b
99b
MC
82
P-P
86
88
71
Memory 63

# TECHNOLOGY TO APPLY DRIVING NORMS FOR AUTOMATED VEHICLE BEHAVIOR PREDICTION

## TECHNICAL FIELD

**[0001]** Embodiments generally relate to automated control systems. More particularly, embodiments relate to technology that learns and applies driving norms in automated vehicle control systems.

## BACKGROUND

**[0002]** Automated control systems may be used in a variety of environments such as, for example, autonomous vehicle environments. Driving a vehicle often requires the interpretation of subtle indirect cues to predict the behavior of other traffic agents. These cues are often relational. Given that the set of allowed (safe) actions a vehicle can execute are limited by the driving agent's ability to communicate, drivers often rely on local driving norms and expected behavior using reasoning and predictability to operate efficiently and safely. The ability to implicitly or explicitly communicate cues helps assure safe driving conditions. While direct interaction between objects in a driving setting poses clear danger, indirect interactions between vehicles and other objects along the road can increase the safety and interpretability of vehicle actions. Drivers gain a considerable amount of information about nearby vehicles based on the adherence of the vehicles (and drivers) to normative driving behavior. For example, indirect interactions between vehicles may communicate the desire to switch lanes, upcoming traffic delays, and more.

**[0003]** Communications between vehicles or between a pedestrian and vehicle is inherently relational as the two agents must exchange information using an agreed upon vocabulary. Deviations from driving norms may present safety challenges for autonomous (i.e., self-driving) vehicles in mixed traffic environments.

## BRIEF DESCRIPTION OF THE DRAWINGS

**[0004]** The various advantages of the embodiments will become apparent to one skilled in the art by reading the following specification and appended claims, and by referencing the following drawings, in which:

**[0005]** FIG. **1** is a diagram illustrating components of an example of an autonomous vehicle system according to one or more embodiments;

**[0006]** FIG. **2** is a block diagram of an example of a relational reasoning system for an autonomous vehicle according to one or more embodiments;

**[0007]** FIG. **3** is a diagram illustrating an example of a graph extraction module of a relational reasoning system according to one or more embodiments;

**[0008]** FIG. **4** is a diagram illustrating an example of a graph attention network of a relational reasoning system according to one or more embodiments;

**[0009]** FIG. **5** is a diagram illustrating an example of a long short-term memory network of a relational reasoning system according to one or more embodiments;

**[0010]** FIG. **6** provides a flowchart illustrating operation of an example of a relational reasoning system for an autonomous vehicle according to one or more embodiments;

**[0011]** FIG. **7** is a block diagram illustrating an example of a performance-enhanced computing system according to one or more embodiments;

**[0012]** FIG. **8** is a block diagram illustrating an example semiconductor apparatus according to one or more embodiments;

**[0013]** FIG. **9** is a block diagram illustrating an example of a processor according to one or more embodiments; and

**[0014]** FIG. **10** is a block diagram illustrating an example of a multiprocessor-based computing system according to one or more embodiments.

## DESCRIPTION OF EMBODIMENTS

**[0015]** In general, embodiments provide a relational reasoning system for an autonomous vehicle that predicts behaviors of traffic participants in a driving environment. Embodiments also provide for efficient prediction of traffic-agents future trajectories and quantification of deviation between observed behavior to predicted behavior for trajectory planning and safety calculations. Additionally, embodiments include technology that will capitalize on relational information and be trained to encode knowledge of driving norms. More particularly, embodiments use a graph attention network to learn relational embeddings which are then fed to a recurrent neural network. The recurrent neural network provides trajectory predictions for an autonomous vehicle as well as for neighboring vehicles and objects, and detects potential collisions.

**[0016]** Embodiments of the relational reasoning system provide autonomous vehicles with the capability of learning and reasoning about regional and local driving behavior to predict intent and improve communication between cars on the road and communication between other individuals such as bikers and pedestrians. Relational communication between agents in a transportation setting relies heavily on adherence to predictable and agreed upon action/responses which can be considered local driving norms. The agent must not only recognize a behavior but also decide if a specific action is communicative. After deciding that an action is meant to communicate an intent, the driving agent must then provide an interpretation for the intent. The same actions in different geographical region and contextual situation might communicate many different things. According to embodiments, the system may quickly generalize to new situations and new locations which have a unique set of norms.

**[0017]** For example, most of the underlying reasoning that supports autonomous vehicles (i.e., self-driving cars) focuses on recognition and trajectory predictions of objects within a particular safety-radius of the self-driving car. While this has been shown to guarantee certain levels of safety, it neglects many of the types of relational information that could also be used to increase safety and predictability of a self-driving system. In the case of indirect communication between two agents, relational information becomes more important than object level information, and communication between drivers is important to road safety. Embodiments use neural network embeddings to learn relational information which can be used for various types of relational reasoning related to self-driving cars, with focus on safety decisions and verification of self-driving cars in terms of extending object detection to infer trajectories of recognized objects and to detect possible collisions, and the resulting implications of collisions or avoidances on the

environment. Such embodiments not only detect objects in the scene, but also reason about how these objects will interact within a constantly changing environment. Additionally, to decrease ambiguity and to increase the amount of computational reasoning a self-driving car can accomplish, embodiments represent normative driving behavior and compare possible indirect communication to normative behavior, by identifying meaningful interactions, considering normative interactions in the specific situation, and comparing the potential deviance from normative behavior to behavioral intent.

**[0018]** FIG. **1** is a diagram illustrating components of an example of an autonomous vehicle system **100** according to one or more embodiments, with reference to components and features described herein including but not limited to the figures and associated description. The autonomous vehicle system **100** may include several modules or subsystems, including a perception module **102**, an environmental module **104**, a planning module **106** and an actuation module **108**. The perception module **102** and the environmental module **104** may collect perceptual features via sensors (e.g. lidar, radar, camera, and location information) and process them to get localization and kinematic information pertaining to relevant agents and objects in the ego vehicle's environment.

**[0019]** This information may be provided as input to the planning module **106**, which may carry out features of the relational reasoning system described in more detail in the following figures. In some embodiments, the planning module **106** may include some or all of components as shown in the breakout illustration in FIG. **1**. The output of planning module **106** may be provided as input to the actuation module **108**, which may carry out actuation commands for controlling steering, acceleration, and/or braking functions of the autonomous vehicle.

**[0020]** FIG. **2** is a block diagram of an example of a relational reasoning system **200** for an autonomous vehicle according to one or more embodiments, with reference to components and features described herein including but not limited to the figures and associated description. Embodiments provide a framework (i.e., subsystem), based on two neural networks, which receives as input processed perceptual features (including, e.g., localization and kinematic information) providing trajectory histories pertaining to the ego vehicle along with other vehicles and objects. The trajectory histories may be converted to graphs by a graph extraction module and fed to a first neural network for driving norm encoding, which in turn may be fed to a second neural network for trajectory prediction. The trajectory prediction may be used to inform actuation commands. The first neural network may be a graph attention (GAT) network to encode driving norms and agent to agent communication with the spatial and temporal information from the driving scene in a relational model. This relational representation may then be provided to the second neural network, which may be a long short-term memory (LSTM) recurrent network, to predict the trajectories of the autonomous vehicle and interacting objects. The GAT-LSTM framework may receive training feedback comparing the predicted to actual trajectories of specific objects interacting within the scope of the autonomous vehicle system. In embodiments, the graph extraction module may be implemented in software executing in a processor, and the GAT and LSTM networks may be implemented in a field programmable gate array (FPGA) accelerator. In this manner, the main part of the model (i.e., the GAT-LSTM) can be trained efficiently in the FPGA, while performing the graph extraction module in the processor can reduce the memory access requirements and the computation to be performed otherwise in the FPGA. In an embodiment, the GAT and LSTM networks may be implemented in a combination of a processor and a FPGA.

**[0021]** During inference, this framework may predict future trajectories and evaluate deviation between predicted trajectories and observed trajectories. The predicted trajectories may include real-time perceptual error information in the calculation of each trajectory, influencing the navigation behavior of the autonomous vehicle. In some embodiments, the predicted trajectories as well as real-time perceptual error information may be paired with safety criteria to provide driving behavior constraints.

**[0022]** As shown in FIG. **2**, a relational reasoning system **200** may include a framework comprising a graph extraction module **210**, a first neural network **220**, and a second neural network **230**. The graph extraction module **210**, as further described with reference to FIG. **3** herein, may generate a series of time-stamped object graphs based on input processed vehicle and object data **240**. The input processed vehicle and object data **240** may be obtained from sensor data (such as, for example, cameras, radar, lidar, etc.), map data, and other data providing information about vehicles and other objects in the vicinity of the ego vehicle, and may be received via a sensor interface **245**. In some embodiments, the input processed vehicle and object data **240** may be obtained from a perception module (e.g., via perception module **102** and/or environmental module **104** as shown in FIG. **1**, already discussed). The perception module may be, e.g., a perception module such as one used in conjunction with the Responsibility-Sensitive Safety (RSS) mathematical framework, introduced by Intel© and Mobileye, for autonomous vehicle operation. Additional data such as indirect interactions between vehicles (e.g., flashing headlights) or between vehicle/pedestrian/biker (e.g., manual turn signal) and other indicators (e.g., turn signals, brake lights, horns, emergency vehicle lights or sirens) may also be included in the input vehicle and object data **240**. In an embodiment, local conditions data **250** may also be input to the graph extraction module **210** and encompassed, along with the processed vehicle and object data, in the generated time-stamped object graphs. The local conditions data **250** may include, for example, one or more of weather conditions, time of day, day of week, day of year, fixed obstacles, etc.

**[0023]** The first neural network **220**, which may be a graph attention (GAT) network as further described with reference to FIG. **4** herein, may receive as input the series of time-stamped object graphs, and learns embeddings that embrace driving norms to generate a series of relational object representations. The second neural network **230**, which may be a long short-term memory (LSTM) recurrent network as further described with reference to FIG. **5** herein, may receive as input the series of relational object representations to determine predicted object trajectories for the ego vehicle and other external objects (including other vehicles). By combining a graph attention network to learn relational and spatial interactions among traffic agents with a long short-term memory network to learn longer term changes and dependencies of each traffic agent through recurrence, this framework leverages both the benefits of relational reason-

ing and that of the temporal sequence learning with neural networks targeted at encoding driving norms to improve trajectory prediction.

**[0024]** The predicted vehicle trajectories **260** (i.e., prediction of future trajectories of the vehicles) resulting from the second neural network **230** may be provided as input to a vehicle navigation actuator subsystem **270** for use in navigating and controlling the autonomous vehicle. Additionally, route planning input **280** from a route planning module and safety criteria input **285** from a safety module may also be applied by the vehicle navigation actuator subsystem **270** in navigating and controlling the autonomous vehicle. Information such as traffic signs, rules of the road (e.g. drive on right side of road, keep right except to pass, pass only if dashed line, etc.) may be utilized by the route planning module to influence route planning input **280**.

**[0025]** FIG. **3** is a diagram **300** illustrating an example of a graph extraction module **310** of a relational reasoning system according to one or more embodiments, with reference to components and features described herein including but not limited to the figures and associated description. The graph extraction module **310** may generally be incorporated into the graph extraction module **210** (FIG. **2**), already discussed. The graph extraction module **310** may receive as input vehicle and object coordinate data **320**. The vehicle and object coordinate data **320**, which may be a vector, may be determined from identified relevant objects and their locations that appeared in the sensor data (e.g., video and/or images). The vehicle and object coordinate data **320** may include, for example, coordinates for the ego vehicle and for other vehicles in the vicinity of the ego vehicle, such as, e.g., other cars, trucks, buses, motorcycles, tractors, etc. These coordinates may be measured at a series of intervals over a particular history time window $\{t_{c-h+1}, \ldots, t_c\}$. In this regard, the vehicle and object coordinate data **320** may represent vehicle and object trajectory histories over the time window of measurement. In some embodiments, the vehicle and object coordinate data **320** may comprise input processed vehicle and object data **240** (FIG. **2**), already discussed. In an embodiment, local conditions data **330**, which may be a vector, may also be input to the graph extraction module **310**. In an embodiment, the local conditions data **330** may comprise the local conditions data **250** (FIG. **2**), already discussed.

**[0026]** The graph extraction module **310** may process the vehicle and object coordinate data **320** by calculating a distance $d_{ij}$ for each pair of objects i and j based on their coordinate values. A graph $G_s=\{V_s, E_s\}$ may then be created for each time point s, where each node in the graph represents an object, and an edge exists between nodes i and j if $d_{ij}<D$, where D is a threshold distance. Once all of the coordinates for the history time window have been processed, the trajectory histories are converted to graphs. That is, the coordinates (object locations/images) at timesteps $\{t_{c-h+1}, \ldots, t_c\}$ are converted to time-stamped graphs $\{G_{t_{c-h+1}}, \ldots, G_{t_c}\}$. Given the output collection of time-stamped object graphs **340** and the coordinate values $(x_{is},y_{is})$ for each node i at each timestamp s, trajectory prediction may be based on predicting the coordinate values for the nodes at future time points $\{t_{c+1}, t_{c+2}, \ldots, t_{c+f}\}$, where f is the size of the future window for which a prediction is to be obtained.

**[0027]** The time-stamped object graphs **340** may be visualized as a time series of two-dimensional graphs **345**, where each plane represents a graph constructed for one of the particular timestamps, and each node in a graph represents an object position. Of course, as constructed the graphs may represent more than two dimensions. For example, each graph generated may encompass three dimensions (representing object position in 3-dimensional space). Graphs of additional dimensions may be generated based on additional input vectors.

**[0028]** FIG. **4** is a diagram **400** illustrating an example of a graph attention network **410** of a relational reasoning system according to one or more embodiments, with reference to components and features described herein including but not limited to the figures and associated description. In general terms, a graph attention network is a neural network that operates on graph-structured data, by stacking neural network layers in which nodes are able to attend to their neighborhoods' features. The graph attention network **410** may generally be incorporated into the first neural network **220** (FIG. **2**), already discussed. The graph attention network **410** is designed to capture the relational interactions among the nodes in the graphs, i.e., the spatial interactions between the traffic agents, which encode information about the driving norm in that geo-location. A set of time-stamped object graphs **420** provides a set of node features (i.e., the coordinate values for each traffic agent) as input to the graph attention network **410**. Each traffic agent is represented as a node in a graph and the edges denote a meaningful relationship between two agents. The relational representation will be encouraged via training on data where interactions between objects are possible and/or communicative such that the model will learn driving norms in diverse environments.

**[0029]** The graph attention network **410** may include a number (M) of stacked neural network layers, and each neural network feed-forward activation layer produces a new set of latent node features, also called embeddings, representing learned relational information. In addition to capturing important relational interactions among nodes, advantages of the graph attention architecture include efficiency in computation, since predictions in graphs can be parallelized and executed independently across node neighborhoods, and inductive learning, i.e., the model can generalize to new/unseen nodes, edges, and graphs.

**[0030]** As illustrated in FIG. **4**, the node embedding for node i in layer L+1 of the graph attention network **410** may be computed from the node features or embeddings of node i and its neighboring nodes N(i) in layer L. Given the node embeddings from layer L, a shared linear transformation, parameterized by a weight matrix W, is applied to each node, and an attentional mechanism (att) is then performed on the nodes to compute the attention coefficients between node i and each neighboring node j:

$$e_{ij}=\mathrm{att}(Wh_i, Wh_j)$$

**[0031]** Each value $e_{ij}$ indicates the importance of node j's features to reference node i. The SoftMax function is used to normalize the attention coefficients across all choices of j:

$$\alpha_{ij} = softmax_j(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{k\in N(i)}\exp(e_{ik})},$$

**[0032]** where node k is a neighbor of node i. In the graph attention network **410**, the attention mechanism att may be

a single-layer feed-forward neural network, parameterized by a learnable weight vector a and applying the LeakyReLU non-linearity. The Leaky Rectified Linear Unit function (LeakyReLU) is an activation function used in neural networks. Fully expanded out, the coefficients computed by the attention mechanism can be expressed as:

$$\alpha_{ij} = \frac{\exp(LeakyReLU(a^T[Wh_i \| Wh_j]))}{\Sigma_{k \in N_i} \exp(LeakyReLU(a^T[Wh_i \| Wh_k]))}$$

**[0033]** As shown in FIG. **4**, node i has neighbors $\{j_1, j_2, j_3, j_4\}$, with their node embeddings $\{h_{j_1}, h_{j_2}, h_{j_3}, h_{j_4}\}$ from layer l. Attention coefficients $\{e_{ij_1}, e_{ij_2}, e_{ij_3}, e_{ij_1}\}$ may be computed, where $e_{ij_1}$=LeakyReLU($a^T[Wh_i \| Wh_{j_1}]$). Then, after applying the SoftMax function, the normalized coefficients $\{\alpha_{ij_1}, \alpha_{ij_2}, \alpha_{ij_3}, \alpha_{ij_4}\}$, may be computed as follows:

$$\alpha_{ij_1} = \frac{\exp(LeakyReLU(a^T[Wh_i \| Wh_{j_1}]))}{\Sigma_{k \in \{j_1, j_2, j_3, j_4\}} \exp(LeakyReLU(a^T[Wh_i \| Wh_k]))},$$

**[0034]** where vectors $\alpha$ and W may be obtained via training. To obtain the (L+1)-layer output embedding $h_i$ for node i, the normalized attention coefficients $\{\alpha_{ij_1}, \alpha_{ij_2}, \alpha_{ij_3}, \alpha_{ij_4}\}$ may then be aggregated via a linear combination of the features of neighboring nodes, and a nonlinearity function $\sigma$ (e.g., Rectified Linear Unit, or ReLU) may be applied:

$$h_i = \sigma(Z_{j \in N(i)} \alpha_{ij} Wh_j)$$

**[0035]** After processing via the M layers of the graph attention network **410**, a resulting set of relational object representations **430** may be obtained. The relational object representations **430** may provide a feature matrix for each time stamp in time window $\{t_{c-h+1}, \ldots, t_c\}$, where each row represents the feature vector for a traffic agent, which has encoded the spatial and communicative interactions between this agent and its neighboring traffic agents. The relational object representations **430** represent learned relationships among the vehicles and other objects over the history time window—including how the relationships vary over the time window.

**[0036]** FIG. **5** is a diagram **500** illustrating an example of a long short-term memory (LSTM) neural network **510** of a relational reasoning system according to one or more embodiments, with reference to components and features described herein including but not limited to the figures and associated description. In general terms, a long short-term memory neural network is a recurrent neural network that incorporates memory cell(s) to make it less sensitive to temporal delay length as compared to other sequence learning models. Thus, in the context of the relational reasoning framework, the LSTM network **510** can process and predict time series given time lags of unknown duration and for graphs of various size and density. Together with the graph attention network **410**, this enables the relational reasoning system to be highly flexible on the duration of history needed for prediction and also the future time period over which the system can predict object trajectories. The LSTM network **510** may generally be incorporated into the second neural network **230** (FIG. **2**), already discussed.

**[0037]** The LTSM network **510** may include an encoder LSTM **520** and a decoder LSTM **530**. Each of the encoder LSTM **520** and the decoder LSTM **530** may itself be a long short-term (LSTM) neural network, where the encoder LSTM is used for encoding the relational representations learned at multiple time points, and the decoder LSTM is adopted for future trajectory prediction. Each of the encoder LSTM **520** and the decoder LSTM **530** maybe a two-layer LSTM network. In some embodiments, the encoder LSTM **520** and/or the decoder LSTM **530** may include an arrangement using three or more layers; the number of layers may be determined to best accommodate the scale and complexity of the collected vehicle data. The relational object representations **540**, the learned relational representations of each traffic agent at each time point together with their temporal features (i.e., information pertaining to local driving norms as output by graph attention network **410**), may be received as input to the LTSM network **510** for encoding, via the encoder LSTM **520**, the temporal location changes of each traffic agent or object. The hidden state of the encoder LSTM **520** and the coordinate values of each agent at the history time points may, in turn, be fed into the decoder LSTM **530** to predict the future trajectories (i.e., object behaviors) of each traffic agent or object, given by the coordinates $Y_{l_{pred}}{}^t = (x_{it}, y_{it})$ for agent i for the future f time points $t = \{t_{c+1}, \ldots, t_{c+f}\}$. The predicted vehicle trajectories **550** (i.e., prediction of future trajectories of the vehicles) may be output from the LSTM network **510** and utilized in connection with the autonomous vehicle actuation, e.g., the vehicle navigation actuator subsystem **270** (FIG. **2**), already discussed. Prediction of object behaviors may include predicting object coordinates (position), orientation (heading) and/or speed attributes (e.g., velocity).

**[0038]** The relational reasoning system (specifically, the graph attention network **410** along with the LSTM network **510**) may be trained using data representing a variety of situations and locations—thus making the relational reasoning system robust and capable of generalizing to changing and variable conditions with geo-location changes and local normative changes. The relational reasoning system GAT-LSTM is an end-to-end framework, and therefore the neural network components in this framework are trained together as a unit. Training data may be obtained from data recordings such as the ones captured in today's automated vehicle fleets. For example, the input to the relational reasoning system may be the output of a perception module at particular times, and the system would be trained based on the accurate prediction of sequential trajectories given the input data. For training purposes, a loss function may be employed to measure error. An error function used to train the system may be based on predicting the future trajectories of traffic agents represented in the training data. As an example, the following mean squared error (MSE) loss function may be used in training the relational reasoning system:

$$\text{Loss} = \frac{1}{f * N} \Sigma_{t=t_{c+1}}^{t_{c+f}} \Sigma_{i=1}^{N} \| Y_{i_{pred}}^t - Y_{i_{true}}^T \|^2,$$

**[0039]** where $t = \{t_{c+1}, t_{c+2} \ldots, t_{c+f}\}$ is the time point in the future, $Y_{i_{pred}}{}^t$ is the predicted coordinate for traffic agent i at time t, and $Y_{i_{true}}{}^t$ is the ground truth (true coordinate for agent i at time t). The relational reasoning system may be trained using a stochastic gradient descent optimizer such as, e.g., the stochastic gradient descent optimizer described in

Kingma, Diederik P., and Jimmy Ba, "Adam: A method for stochastic optimization," available via arXiv preprint: arXiv: 1412.6980 (2014).

**[0040]** FIG. **6** provides a flowchart illustrating a process **600** for operating an example of a relational reasoning system for an autonomous vehicle according to one or more embodiments, with reference to components and features described herein including but not limited to the figures and associated description. Process **600** may be implemented in relational reasoning system **200** described herein with reference to FIG. **2**, already discussed. More particularly, the process **600** may be implemented in one or more modules as a set of logic instructions stored in a machine- or computer-readable storage medium such as random access memory (RAM), read only memory (ROM), programmable ROM (PROM), firmware, flash memory, etc., in configurable logic such as, for example, programmable logic arrays (PLAs), field programmable gate arrays (FPGAs), complex programmable logic devices (CPLDs), in fixed-functionality logic hardware using circuit technology such as, for example, application specific integrated circuit (ASIC), complementary metal oxide semiconductor (CMOS) or transistor-transistor logic (TTL) technology, or any combination thereof.

**[0041]** For example, computer program code to carry out operations shown in process **600** may be written in any combination of one or more programming languages, including an object oriented programming language such as JAVA, SMALLTALK, C++ or the like and conventional procedural programming languages, such as the "C" programming language or similar programming languages. Additionally, logic instructions might include assembler instructions, instruction set architecture (ISA) instructions, machine instructions, machine dependent instructions, microcode, state-setting data, configuration data for integrated circuitry, state information that personalizes electronic circuitry and/or other structural components that are native to hardware (e.g., host processor, central processing unit/CPU, microcontroller, etc.).

**[0042]** Illustrated processing block **610** provides for generating a series of time-stamped object graphs based on object trajectory histories derived from external object data for a plurality of external objects. The external object data may include the vehicle and processed vehicle and object data **240** (FIG. **2**) or the object coordinate data **320** (FIG. **3**), already discussed. The series of time-stamped object graphs based on object trajectory histories may be generated via the graph extraction module **310** (FIG. **3**), already discussed, and may include the time-stamped object graphs **340** (FIG. **3**), already discussed.

**[0043]** Illustrated processing block **620** provides for generating, via a first neural network, a series of relational object representations based on the series of time-stamped object graphs. The first neural network may include the neural network **220** (FIG. **2**) or the graph attention network **410** (FIG. **4**), already discussed. The series of relational object representations may include the relational object representations **430**, already discussed.

**[0044]** Illustrated processing block **630** provides for determining, via a second neural network, a prediction of future object trajectories for the plurality of external objects based on the series of relational object representations. The second neural network may include the neural network **230** (FIG. **2**) or the LSTM network **510** (FIG. **5**), already discussed. The prediction of future object trajectories for the plurality of external objects may include the predicted vehicle trajectories **260** (FIG. **2**) or the predicted vehicle trajectories **550** (FIG. **5**), already discussed.

**[0045]** The predicted object trajectories for the plurality of external objects (block **630**) may be used by an autonomous vehicle for navigation purposes. For example, illustrated processing block **640** provides for including real-time perceptual error information with the predicted object trajectories. Next, illustrated processing block **650** provides for modifying the vehicle behavior based on the predicted object trajectories and real-time perceptual error information. Modifying vehicle behavior may include issuing actuation commands to navigate the vehicle. Actuation commands may be different depending on the low-level controller of the vehicle. In general, the low-level controller is given a reference target speed and a path composed of a sequence of points in the vehicle reference frame that the controller seeks to adhere to. That is, the controller sets the steering wheel and throttle/brake to maintain that target speed while going to the next points that compose the path. In some embodiments, actuation commands may include values for throttle, braking and steering angle.

**[0046]** In some embodiments, the predicted trajectories as well as real-time perceptual error information may be paired with safety criteria to provide driving behavior constraints. Safety criteria may generally be understood to include rules or guidelines for collision avoidance, for example by establishing a minimum longitudinal and lateral distance metric during a particular situation. Safety criteria may also include local rules of the road such as maximum speed in the road segment, respecting signals, and/or allowing—or prohibiting—certain manoeuvres (e.g., at intersections). To help ensure safety, the predicted object trajectories for the plurality of external objects (block **630**) may also be used by an autonomous vehicle to modify or constrain vehicle behavior even more than provided by safety criteria. For example, illustrated processing block **660** provides for determining the deviation of observed object behaviors from predicted object behaviors. Next, illustrated processing block **670** provides for modifying the vehicle behavior based on the determined deviation of object behavior from predicted behavior. Examples of modifying the ego vehicle behavior may include: 1) increasing longitudinal distance to another vehicle in the same lane and direction, 2) increasing minimum lateral distance to a road user in an adjacent lane, 3) giving way to another vehicle at an intersection (even if the ego vehicle has priority or right-of-way, 4) reducing current speed (e.g., in areas with occlusion or other obstacles) even if speed is within the maximum speed allowed for the current road segment.

**[0047]** FIG. **7** shows a block diagram illustrating an example computing system **10** for predicting vehicle trajectories based on local driving norms according to one or more embodiments, with reference to components and features described herein including but not limited to the figures and associated description. The system **10** may generally be part of an electronic device/platform having computing and/or communications functionality (e.g., server, cloud infrastructure controller, database controller, notebook computer, desktop computer, personal digital assistant/PDA, tablet computer, convertible tablet, smart phone, etc.), imaging functionality (e.g., camera, camcorder), media playing functionality (e.g., smart television/TV), wearable functionality (e.g., watch, eyewear, headwear, footwear, jewelry), vehicu-

lar functionality (e.g., car, truck, motorcycle), robotic functionality (e.g., autonomous robot), Internet of Things (IoT) functionality, etc., or any combination thereof. In the illustrated example, the system **10** may include a host processor **12** (e.g., central processing unit/CPU) having an integrated memory controller (IMC) **14** that may be coupled to system memory **20**. The host processor **12** may include any type of processing device, such as, e.g., microcontroller, microprocessor, RISC processor, ASIC, etc., along with associated processing modules or circuitry. The system memory **20** may include any non-transitory machine- or computer-readable storage medium such as RAM, ROM, PROM, EEPROM, firmware, flash memory, etc., configurable logic such as, for example, PLAs, FPGAs, CPLDs, fixed-functionality hardware logic using circuit technology such as, for example, ASIC, CMOS or TTL technology, or any combination thereof suitable for storing instructions **28**.

**[0048]** The system **10** may also include an input/output (I/O) subsystem **16**. The IO subsystem **16** may communicate with for example, one or more input/output (I/O) devices **17**, a network controller **24** (e.g., wired and/or wireless NIC), and storage **22**. The storage **22** may be comprised of any appropriate non-transitory machine- or computer-readable memory type (e.g., flash memory, DRAM, SRAM (static random access memory), solid state drive (SSD), hard disk drive (HDD), optical disk, etc.). The storage **22** may include mass storage. In some embodiments, the host processor **12** and/or the I/O subsystem **16** may communicate with the storage **22** (all or portions thereof) via the network controller **24**. In some embodiments, the system **10** may also include a graphics processor **26** (e.g., graphics processing unit/GPU) and an AI accelerator **27**. In some embodiments, the system **10** may also include a perception subsystem **18** (e.g., including one or more sensors and/or cameras) and/or an actuation subsystem **19**. In an embodiment, the system **10** may also include a vision processing unit (VPU), not shown.

**[0049]** The host processor **12** and the I/O subsystem **16** may be implemented together on a semiconductor die as a system on chip (SoC) **11**, shown encased in a solid line. The SoC **11** may therefore operate as a computing apparatus for autonomous vehicle control. In some embodiments, the SoC **11** may also include one or more of the system memory **20**, the network controller **24**, the graphics processor **26** and/or the AI accelerator **27** (shown encased in dotted lines). In some embodiments, SoC **11** may also include other components of system **10**.

**[0050]** The host processor **12**, the I/O subsystem **16**, the graphics processor **26**, the Al accelerator **27** and/or the VPU may execute program instructions **28** retrieved from the system memory **20** and/or the storage **22** to perform one or more aspects of process **600** as described herein with reference to FIG. **6**. Thus, execution of instructions **28** may cause the SoC **11** to generate a series of time-stamped object graphs based on object trajectory histories derived from external object data for a plurality of external objects, generate, via a first neural network, a series of relational object representations based on the series of time-stamped object graphs, and determine, via a second neural network, predicted object trajectories for the plurality of external objects based on the series of relational object representations. The system **10** may implement one or more aspects of the autonomous vehicle system **100**, the relational reasoning system **200**, the graph extraction module **310**, the graph attention network **410**, and/or the LSTM network **510** as described herein with reference to FIGS. **1-5**. The system **10** is therefore considered to be performance-enhanced at least to the extent that vehicle and object trajectories may be predicted based on local driving norms.

**[0051]** Computer program code to carry out the processes described above may be written in any combination of one or more programming languages, including an object-oriented programming language such as JAVA, JAVASCRIPT, PYTHON, SMALLTALK, C++ or the like and/or conventional procedural programming languages, such as the "C" programming language or similar programming languages, and implemented as program instructions **28**. Additionally, program instructions **28** may include assembler instructions, instruction set architecture (ISA) instructions, machine instructions, machine dependent instructions, microcode, state-setting data, configuration data for integrated circuitry, state information that personalizes electronic circuitry and/or other structural components that are native to hardware (e.g., host processor, central processing unit/CPU, microcontroller, microprocessor, etc.).

**[0052]** The I/O devices **17** may include one or more of input devices, such as a touch-screen, keyboard, mouse, cursor-control device, touch-screen, microphone, digital camera, video recorder, camcorder, biometric scanners and/or sensors; input devices may be used to enter information and interact with system **10** and/or with other devices. The I/O devices **17** may also include one or more of output devices, such as a display (e.g., touch screen, liquid crystal display/LCD, light emitting diode/LED display, plasma panels, etc.), speakers and/or other visual or audio output devices. Input and/or output devices may be used, e.g., to provide a user interface.

**[0053]** FIG. **8** shows a block diagram illustrating an example semiconductor apparatus **30** for predicting vehicle trajectories based on local driving norms according to one or more embodiments, with reference to components and features described herein including but not limited to the figures and associated description. The semiconductor apparatus **30** may be implemented, e.g., as a chip, die, or other semiconductor package. The semiconductor apparatus **30** may include one or more substrates **32** comprised of, e.g., silicon, sapphire, gallium arsenide, etc. The semiconductor apparatus **30** may also include logic **34** comprised of, e.g., transistor array(s) and other integrated circuit (IC) components) coupled to the substrate(s) **32**. The logic **34** may be implemented at least partly in configurable logic or fixed-functionality logic hardware. The logic **34** may implement system on chip (SoC) **11** described above with reference to FIG. **7**. The logic **34** may implement one or more aspects of process **600** as described herein with reference to FIG. **6**, including generate a series of time-stamped object graphs based on object trajectory histories derived from external object data for a plurality of external objects, generate, via a first neural network, a series of relational object representations based on the series of time-stamped object graphs, and determine, via a second neural network, predicted object trajectories for the plurality of external objects based on the series of relational object representations. The logic **34** may implement one or more aspects of the autonomous vehicle system **100**, the relational reasoning system **200**, the graph extraction module **310**, the graph attention network **410**, and/or the LSTM network **510** as described herein with reference to FIGS. **1-5**. The apparatus **30** is therefore con-

sidered to be performance-enhanced at least to the extent that vehicle and object trajectories may be predicted based on local driving norms.

**[0054]** The semiconductor apparatus **30** may be constructed using any appropriate semiconductor manufacturing processes or techniques. For example, the logic **34** may include transistor channel regions that are positioned (e.g., embedded) within the substrate(s) **32**. Thus, the interface between the logic **34** and the substrate(s) **32** may not be an abrupt junction. The logic **34** may also be considered to include an epitaxial layer that is grown on an initial wafer of the substrate(s) **34**.

**[0055]** FIG. **9** is a block diagram illustrating an example processor core **40** according to one or more embodiments, with reference to components and features described herein including but not limited to the figures and associated description. The processor core **40** may be the core for any type of processor, such as a micro-processor, an embedded processor, a digital signal processor (DSP), a network processor, or other device to execute code. Although only one processor core **40** is illustrated in FIG. **9**, a processing element may alternatively include more than one of the processor core **40** illustrated in FIG. **9**. The processor core **40** may be a single-threaded core or, for at least one embodiment, the processor core **40** may be multithreaded in that it may include more than one hardware thread context (or "logical processor") per core.

**[0056]** FIG. **9** also illustrates a memory **41** coupled to processor core **40**. The memory **41** may be any of a wide variety of memories (including various layers of memory hierarchy) as are known or otherwise available to those of skill in the art. The memory **41** may include one or more code **42** instruction(s) to be executed by the processor core **40**. The code **42** may implement one or more aspects of the process **600** as described herein with reference to FIG. **6**. The processor core **40** may implement one or more aspects of the autonomous vehicle system **100**, the relational reasoning system **200**, the graph extraction module **310**, the graph attention network **410**, and/or the LSTM network **510** as described herein with reference to FIGS. **1-5**. The processor core **40** follows a program sequence of instructions indicated by the code **42**. Each instruction may enter a front end portion **43** and be processed by one or more decoders **44**. The decoder **44** may generate as its output a micro operation such as a fixed width micro operation in a predefined format, or may generate other instructions, microinstructions, or control signals which reflect the original code instruction. The illustrated front end portion **43** also includes register renaming logic **46** and scheduling logic **48**, which generally allocate resources and queue the operation corresponding to the convert instruction for execution.

**[0057]** The processor core **40** is shown including execution logic **50** having a set of execution units **55-1** through **55**-N. Some embodiments may include a number of execution units dedicated to specific functions or sets of functions. Other embodiments may include only one execution unit or one execution unit that can perform a particular function. The illustrated execution logic **50** performs the operations specified by code instructions.

**[0058]** After completion of execution of the operations specified by the code instructions, back end logic **58** retires the instructions of the code **42**. In one embodiment, the processor core **40** allows out of order execution but requires in order retirement of instructions. The retirement logic **59** may take a variety of forms as known to those of skill in the art (e.g., re-order buffers or the like). In this manner, the processor core **40** is transformed during execution of the code **42**, at least in terms of the output generated by the decoder, the hardware registers and tables utilized by the register renaming logic **46**, and any registers (not shown) modified by the execution logic **50**.

**[0059]** Although not illustrated in FIG. **9**, a processing element may include other elements on chip with the processor core **40**. For example, a processing element may include memory control logic along with the processor core **40**. The processing element may include I/O control logic and/or may include I/O control logic integrated with memory control logic. The processing element may also include one or more caches.

**[0060]** FIG. **10** is a block diagram illustrating an example of a multi-processor based computing system **60** according to one or more embodiments, with reference to components and features described herein including but not limited to the figures and associated description. The multiprocessor system **60** includes a first processing element **70** and a second processing element **80**. While two processing elements **70** and **80** are shown, it is to be understood that an embodiment of the system **60** may also include only one such processing element.

**[0061]** The system **60** is illustrated as a point-to-point interconnect system, wherein the first processing element **70** and the second processing element **80** are coupled via a point-to-point interconnect **71**. It should be understood that any or all of the interconnects illustrated in FIG. **10** may be implemented as a multi-drop bus rather than point-to-point interconnect.

**[0062]** As shown in FIG. **10**, each of processing elements **70** and **80** may be multicore processors, including first and second processor cores (i.e., processor cores **74***a* and **74***b* and processor cores **84***a* and **84***b*). Such cores **74***a*, **74***b*, **84***a*, **84***b* may be configured to execute instruction code in a manner similar to that discussed above in connection with FIG. **9**.

**[0063]** Each processing element **70**, **80** may include at least one shared cache **99***a*, **99***b*. The shared cache **99***a*, **99***b* may store data (e.g., instructions) that are utilized by one or more components of the processor, such as the cores **74***a*, **74***b* and **84***a*, **84***b*, respectively. For example, the shared cache **99***a*, **99***b* may locally cache data stored in a memory **62**, **63** for faster access by components of the processor. In one or more embodiments, the shared cache **99***a*, **99***b* may include one or more mid-level caches, such as level 2 (L2), level 3 (L3), level 4 (L4), or other levels of cache, a last level cache (LLC), and/or combinations thereof.

**[0064]** While shown with only two processing elements **70**, **80**, it is to be understood that the scope of the embodiments are not so limited. In other embodiments, one or more additional processing elements may be present in a given processor. Alternatively, one or more of processing elements **70**, **80** may be an element other than a processor, such as an accelerator or a field programmable gate array. For example, additional processing element(s) may include additional processors(s) that are the same as a first processor **70**, additional processor(s) that are heterogeneous or asymmetric to processor a first processor **70**, accelerators (such as, e.g., graphics accelerators or digital signal processing (DSP) units), field programmable gate arrays, or any other processing element. There can be a variety of differences between

the processing elements **70**, **80** in terms of a spectrum of metrics of merit including architectural, micro architectural, thermal, power consumption characteristics, and the like. These differences may effectively manifest themselves as asymmetry and heterogeneity amongst the processing elements **70**, **80**. For at least one embodiment, the various processing elements **70**, **80** may reside in the same die package.

**[0065]** The first processing element **70** may further include memory controller logic (MC) **72** and point-to-point (P-P) interfaces **76** and **78**. Similarly, the second processing element **80** may include a MC **82** and P-P interfaces **86** and **88**. As shown in FIG. **10**, the MC's **72** and **82** couple the processors to respective memories, namely a memory **62** and a memory **63**, which may be portions of main memory locally attached to the respective processors. While the MC **72** and **82** is illustrated as integrated into the processing elements **70**, **80**, for alternative embodiments the MC logic may be discrete logic outside the processing elements **70**, **80** rather than integrated therein.

**[0066]** The first processing element **70** and the second processing element **80** may be coupled to an I/O subsystem **90** via P-P interconnects **76** and **86**, respectively. As shown in FIG. **10**, the I/O subsystem **90** includes P-P interfaces **94** and **98**. Furthermore, the I/O subsystem **90** includes an interface **92** to couple the I/O subsystem **90** with a high performance graphics engine **64**. In one embodiment, bus **73** may be used to couple the graphics engine **64** to the I/O subsystem **90**. Alternately, a point-to-point interconnect may couple these components.

**[0067]** In turn, the I/O subsystem **90** may be coupled to a first bus **65** via an interface **96**. In one embodiment, the first bus **65** may be a Peripheral Component Interconnect (PCI) bus, or a bus such as a PCI Express bus or another third generation I/O interconnect bus, although the scope of the embodiments are not so limited.

**[0068]** As shown in FIG. **10**, various I/O devices **65***a* (e.g., biometric scanners, speakers, cameras, sensors) may be coupled to the first bus **65**, along with a bus bridge **66** which may couple the first bus **65** to a second bus **67**. In one embodiment, the second bus **67** may be a low pin count (LPC) bus. Various devices may be coupled to the second bus **67** including, for example, a keyboard/mouse **67***a*, communication device(s) **67***b*, and a data storage unit **68** such as a disk drive or other mass storage device which may include code **69**, in one embodiment. The illustrated code **69** may implement one or more aspects of the process **600** as described herein with reference to FIG. **6**. The illustrated code **69** may be similar to code **42** (FIG. **9**), already discussed. Further, an audio I/O **67***c* may be coupled to second bus **67** and a battery **61** may supply power to the computing system **60**. The system **60** may implement one or more aspects of the autonomous vehicle system **100**, the relational reasoning system **200**, the graph extraction module **310**, the graph attention network **410**, and/or the LSTM network **510** as described herein with reference to FIGS. **1-5**.

**[0069]** Note that other embodiments are contemplated. For example, instead of the point-to-point architecture of FIG. **10**, a system may implement a multi-drop bus or another such communication topology. Also, the elements of FIG. **10** may alternatively be partitioned using more or fewer integrated chips than shown in FIG. **10**.

**[0070]** Embodiments of each of the above systems, devices, components and/or methods, including the system **10**, the semiconductor apparatus **30**, the processor core **40**, the system **60**, the autonomous vehicle system **100**, the relational reasoning system **200**, the graph extraction module **310**, the graph attention network **410**, the LSTM network **510**, and/or the process **600**, and/or any other system components, may be implemented in hardware, software, or any suitable combination thereof. For example, hardware implementations may include configurable logic such as, for example, programmable logic arrays (PLAs), field programmable gate arrays (FPGAs), complex programmable logic devices (CPLDs), or fixed-functionality logic hardware using circuit technology such as, for example, application specific integrated circuit (ASIC), complementary metal oxide semiconductor (CMOS) or transistor-transistor logic (TTL) technology, or any combination thereof.

**[0071]** Alternatively, or additionally, all or portions of the foregoing systems and/or components and/or methods may be implemented in one or more modules as a set of logic instructions stored in a machine- or computer-readable storage medium such as random access memory (RAM), read only memory (ROM), programmable ROM (PROM), firmware, flash memory, etc., to be executed by a processor or computing device. For example, computer program code to carry out the operations of the components may be written in any combination of one or more operating system (OS) applicable/appropriate programming languages, including an object-oriented programming language such as PYTHON, PERL, JAVA, SMALLTALK, C++, C# or the like and conventional procedural programming languages, such as the "C" programming language or similar programming languages.

## Additional Notes and Examples

**[0072]** Example 1 includes a computing system comprising a sensor interface to receive external object data, a processor coupled to the sensor interface, the processor including one or more substrates and logic coupled to the one or more substrates, wherein the logic is implemented at least partly in one or more of configurable logic or fixed-functionality hardware logic, the logic coupled to the one or more substrates to generate a series of time-stamped object graphs based on object trajectory histories derived from the external object data for a plurality of external objects, generate, via a first neural network, a series of relational object representations based on the series of time-stamped object graphs, and determine, via a second neural network, a prediction of future object trajectories for the plurality of external objects based on the series of relational object representations.

**[0073]** Example 2 includes the system of Example 1, wherein the logic coupled to the one or more substrates is further to include real-time perceptual error information with the predicted object trajectories, and modify behavior of an autonomous vehicle based on the predicted object trajectories and the real-time perceptual error information.

**[0074]** Example 3 includes the system of Example 1, wherein the logic coupled to the one or more substrates is further to determine deviation of observed object behaviors from predicted object behaviors, and modify behavior of an autonomous vehicle based on the determined object behavioral deviation.

**[0075]** Example 4 includes the system of Example 1, wherein the object trajectory histories include coordinates for a plurality of vehicles within a time window, wherein the series of time-stamped object graphs assist learning how the vehicles relate over the time window, wherein the relational object representations represent learned relationships among the plurality of vehicles over the time window, and wherein the first neural network is to encode location-based driving norms.

**[0076]** Example 5 includes the system of Example 4, wherein the second neural network comprises a first recurrent neural network that is to encode temporal vehicle location changes and a second recurrent neural network that is to predict future behaviors for the plurality of vehicles.

**[0077]** Example 6 includes the system of any of Examples 1-5, wherein the first neural network comprises a graph attention (GAT) network and the second neural network comprises a long short-term memory (LSTM) network, and wherein the first neural network and the second neural network are trained as a unit using object trajectory histories generated from relational object data obtained from vehicle driving data collected across a plurality of geographic locations.

**[0078]** Example 7 includes a semiconductor apparatus comprising one or more substrates, and logic coupled to the one or more substrates, wherein the logic is implemented at least partly in one or more of configurable logic or fixed-functionality hardware logic, the logic coupled to the one or more substrates to generate a series of time-stamped object graphs based on object trajectory histories derived from external object data for a plurality of external objects, generate, via a first neural network, a series of relational object representations based on the series of time-stamped object graphs, and determine, via a second neural network, a prediction of future object trajectories for the plurality of external objects based on the series of relational object representations.

**[0079]** Example 8 includes the semiconductor apparatus of Example 7, wherein the logic coupled to the one or more substrates is further to include real-time perceptual error information with the predicted object trajectories, and modify behavior of an autonomous vehicle based on the predicted object trajectories and the real-time perceptual error information.

**[0080]** Example 9 includes the semiconductor apparatus of Example 7, wherein the logic coupled to the one or more substrates is further to determine deviation of observed object behaviors from predicted object behaviors, and modify behavior of an autonomous vehicle based on the determined object behavioral deviation.

**[0081]** Example 10 includes the semiconductor apparatus of Example 7, wherein the object trajectory histories include coordinates for a plurality of vehicles within a time window, wherein the series of time-stamped object graphs assist learning how the vehicles relate over the time window, wherein the relational object representations represent learned relationships among the plurality of vehicles over the time window, and wherein the first neural network is to encode location-based driving norms.

**[0082]** Example 11 includes the semiconductor apparatus of Example 10, wherein the second neural network comprises a first recurrent neural network that is to encode temporal vehicle location changes and a second recurrent neural network that is to predict future behaviors for the plurality of vehicles.

**[0083]** Example 12 includes the semiconductor apparatus of any of Examples 7-11, wherein the first neural network comprises a graph attention (GAT) network and the second neural network comprises a long short-term memory (LSTM) network, and wherein the first neural network and the second neural network are trained as a unit using object trajectory histories generated from relational object data obtained from vehicle driving data collected across a plurality of geographic locations.

**[0084]** Example 13 includes the semiconductor apparatus of Example 7, wherein the logic coupled to the one or more substrates includes transistor channel regions that are positioned within the one or more substrates.

**[0085]** Example 14 includes at least one non-transitory computer readable storage medium comprising a set of instructions which, when executed by a computing system, cause the computing system to generate a series of time-stamped object graphs based on object trajectory histories derived from external object data for a plurality of external objects, generate, via a first neural network, a series of relational object representations based on the series of time-stamped object graphs, and determine, via a second neural network, a prediction of future object trajectories for the plurality of external objects based on the series of relational object representations.

**[0086]** Example 15 includes the at least one non-transitory computer readable storage medium of Example 14, wherein the instructions, when executed, further cause the computing system to include real-time perceptual error information with the predicted object trajectories, and modify behavior of an autonomous vehicle based on the predicted object trajectories and the real-time perceptual error information.

**[0087]** Example 16 includes the at least one non-transitory computer readable storage medium of Example 14, wherein the instructions, when executed, further cause the computing system to determine deviation of observed object behaviors from predicted object behaviors, and modify behavior of an autonomous vehicle based on the determined object behavioral deviation.

**[0088]** Example 17 includes the at least one non-transitory computer readable storage medium of Example 14, wherein the object trajectory histories include coordinates for a plurality of vehicles within a time window, wherein the series of time-stamped object graphs assist learning how the vehicles relate over the time window, wherein the relational object representations represent learned relationships among the plurality of vehicles over the time window, and wherein the first neural network is to encode location-based driving norms.

**[0089]** Example 18 includes the at least one non-transitory computer readable storage medium of Example 17, wherein the second neural network comprises a first recurrent neural network that is to encode temporal vehicle location changes and a second recurrent neural network that is to predict future behaviors for the plurality of vehicles.

**[0090]** Example 19 includes the at least one non-transitory computer readable storage medium of any of Examples 14-18, wherein the first neural network comprises a graph attention (GAT) network and the second neural network comprises a long short-term memory (LSTM) network, and wherein the first neural network and the second neural

network are trained as a unit using object trajectory histories generated from relational object data obtained from vehicle driving data collected across a plurality of geographic locations.

**[0091]** Example 20 includes a relational reasoning method comprising generating a series of time-stamped object graphs based on object trajectory histories derived from external object data for a plurality of external objects, generating, via a first neural network, a series of relational object representations based on the series of time-stamped object graphs, and determining, via a second neural network, a prediction of future object trajectories for the plurality of external objects based on the series of relational object representations.

**[0092]** Example 21 includes the method of Example 20, further comprising including real-time perceptual error information with the predicted object trajectories, and modifying behavior of an autonomous vehicle based on the predicted object trajectories and the real-time perceptual error information.

**[0093]** Example 22 includes the method of Example 20, further comprising determining deviation of observed object behaviors from predicted object behaviors, and modifying behavior of an autonomous vehicle based on the determined object behavioral deviation.

**[0094]** Example 23 includes the method of Example 20, wherein the object trajectory histories include coordinates for a plurality of vehicles within a time window, wherein the series of time-stamped object graphs assist learning how the vehicles relate over the time window, wherein the relational object representations represent learned relationships among the plurality of vehicles over the time window, and wherein the first neural network encodes location-based driving norms.

**[0095]** Example 24 includes the method of Example 23, wherein the second neural network comprises a first recurrent neural network that encodes temporal vehicle location changes and a second recurrent neural network that predicts future behaviors for the plurality of vehicles.

**[0096]** Example 25 includes the method of any of Examples 20-24, wherein the first neural network comprises a graph attention (GAT) network and the second neural network comprises a long short-term memory (LSTM) network, and wherein the first neural network and the second neural network are trained as a unit using object trajectory histories generated from relational object data obtained from vehicle driving data collected across a plurality of geographic locations.

**[0097]** Example 26 includes an apparatus comprising means for performing the method of any of Examples 20-24.

**[0098]** Thus, technology described herein provides for efficient and robust prediction of future trajectories for an autonomous vehicle as well as for neighboring vehicles and objects by generalizing social driving norms and other types of relational information. The technology prioritizes actions and responses based on relational cues from the driving environment including geo-spatial information about standard driving norms. Additionally, the technology enables navigating the vehicle based on predicted object trajectories and real-time perceptual error information, and modifying safety criteria based on deviation of object behavior from predicted behavior.

**[0099]** Embodiments are applicable for use with all types of semiconductor integrated circuit ("IC") chips. Examples of these IC chips include but are not limited to processors, controllers, chipset components, programmable logic arrays (PLAs), memory chips, network chips, systems on chip (SoCs), SSD/NAND controller ASICs, and the like. In addition, in some of the drawings, signal conductor lines are represented with lines. Some may be different, to indicate more constituent signal paths, have a number label, to indicate a number of constituent signal paths, and/or have arrows at one or more ends, to indicate primary information flow direction. This, however, should not be construed in a limiting manner. Rather, such added detail may be used in connection with one or more exemplary embodiments to facilitate easier understanding of a circuit. Any represented signal lines, whether or not having additional information, may actually comprise one or more signals that may travel in multiple directions and may be implemented with any suitable type of signal scheme, e.g., digital or analog lines implemented with differential pairs, optical fiber lines, and/or single-ended lines.

**[0100]** Example sizes/models/values/ranges may have been given, although embodiments are not limited to the same. As manufacturing techniques (e.g., photolithography) mature over time, it is expected that devices of smaller size could be manufactured. In addition, well known power/ground connections to IC chips and other components may or may not be shown within the figures, for simplicity of illustration and discussion, and so as not to obscure certain aspects of the embodiments. Further, arrangements may be shown in block diagram form in order to avoid obscuring embodiments, and also in view of the fact that specifics with respect to implementation of such block diagram arrangements are highly dependent upon the computing system within which the embodiment is to be implemented, i.e., such specifics should be well within purview of one skilled in the art. Where specific details (e.g., circuits) are set forth in order to describe example embodiments, it should be apparent to one skilled in the art that embodiments can be practiced without, or with variation of, these specific details. The description is thus to be regarded as illustrative instead of limiting.

**[0101]** The term "coupled" may be used herein to refer to any type of relationship, direct or indirect, between the components in question, and may apply to electrical, mechanical, fluid, optical, electromagnetic, electromechanical or other connections. In addition, the terms "first", "second", etc. may be used herein only to facilitate discussion, and carry no particular temporal or chronological significance unless otherwise indicated.

**[0102]** As used in this application and in the claims, a list of items joined by the term "one or more of" may mean any combination of the listed terms. For example, the phrases "one or more of A, B or C" may mean A; B; C; A and B; A and C; B and C; or A, B and C.

**[0103]** Those skilled in the art will appreciate from the foregoing description that the broad techniques of the embodiments can be implemented in a variety of forms. Therefore, while the embodiments have been described in connection with particular examples thereof, the true scope of the embodiments should not be so limited since other modifications will become apparent to the skilled practitioner upon a study of the drawings, specification, and following claims.

We claim:

1. A computing system comprising:

a sensor interface to receive external object data; and

a processor coupled to the sensor interface, the processor including one or more substrates and logic coupled to the one or more substrates, wherein the logic is implemented at least partly in one or more of configurable logic or fixed-functionality hardware logic, the logic coupled to the one or more substrates to:

generate a series of time-stamped object graphs based on object trajectory histories derived from the external object data for a plurality of external objects;

generate, via a first neural network, a series of relational object representations based on the series of time-stamped object graphs; and

determine, via a second neural network, a prediction of future object trajectories for the plurality of external objects based on the series of relational object representations.

2. The system of claim 1, wherein the logic coupled to the one or more substrates is further to:

include real-time perceptual error information with the predicted object trajectories; and

modify behavior of an autonomous vehicle based on the predicted object trajectories and the real-time perceptual error information.

3. The system of claim 1, wherein the logic coupled to the one or more substrates is further to:

determine deviation of observed object behaviors from predicted object behaviors; and

modify behavior of an autonomous vehicle based on the determined object behavioral deviation.

4. The system of claim 1, wherein the object trajectory histories include coordinates for a plurality of vehicles within a time window, wherein the series of time-stamped object graphs assist learning how the vehicles relate over the time window, wherein the relational object representations represent learned relationships among the plurality of vehicles over the time window, and wherein the first neural network is to encode location-based driving norms.

5. The system of claim 4, wherein the second neural network comprises a first recurrent neural network that is to encode temporal vehicle location changes and a second recurrent neural network that is to predict future behaviors for the plurality of vehicles.

6. The system of claim 5, wherein the first neural network comprises a graph attention (GAT) network and the second neural network comprises a long short-term memory (LSTM) network, and wherein the first neural network and the second neural network are trained as a unit using object trajectory histories generated from relational object data obtained from vehicle driving data collected across a plurality of geographic locations.

7. A semiconductor apparatus comprising:

one or more substrates; and

logic coupled to the one or more substrates, wherein the logic is implemented at least partly in one or more of configurable logic or fixed-functionality hardware logic, the logic coupled to the one or more substrates to:

generate a series of time-stamped object graphs based on object trajectory histories derived from external object data for a plurality of external objects;

generate, via a first neural network, a series of relational object representations based on the series of time-stamped object graphs; and

determine, via a second neural network, a prediction of future object trajectories for the plurality of external objects based on the series of relational object representations.

8. The semiconductor apparatus of claim 7, wherein the logic coupled to the one or more substrates is further to:

include real-time perceptual error information with the predicted object trajectories; and

modify behavior of an autonomous vehicle based on the predicted object trajectories and the real-time perceptual error information.

9. The semiconductor apparatus of claim 7, wherein the logic coupled to the one or more substrates is further to:

determine deviation of observed object behaviors from predicted object behaviors; and

modify behavior of an autonomous vehicle based on the determined object behavioral deviation.

10. The semiconductor apparatus of claim 7, wherein the object trajectory histories include coordinates for a plurality of vehicles within a time window, wherein the series of time-stamped object graphs assist learning how the vehicles relate over the time window, wherein the relational object representations represent learned relationships among the plurality of vehicles over the time window, and wherein the first neural network is to encode location-based driving norms.

11. The semiconductor apparatus of claim 10, wherein the second neural network comprises a first recurrent neural network that is to encode temporal vehicle location changes and a second recurrent neural network that is to predict future behaviors for the plurality of vehicles.

12. The semiconductor apparatus of claim 11, wherein the first neural network comprises a graph attention (GAT) network and the second neural network comprises a long short-term memory (LSTM) network, and wherein the first neural network and the second neural network are trained as a unit using object trajectory histories generated from relational object data obtained from vehicle driving data collected across a plurality of geographic locations.

13. The semiconductor apparatus of claim 7, wherein the logic coupled to the one or more substrates includes transistor channel regions that are positioned within the one or more substrates.

14. At least one non-transitory computer readable storage medium comprising a set of instructions which, when executed by a computing system, cause the computing system to:

generate a series of time-stamped object graphs based on object trajectory histories derived from external object data for a plurality of external objects;

generate, via a first neural network, a series of relational object representations based on the series of time-stamped object graphs; and

determine, via a second neural network, a prediction of future object trajectories for the plurality of external objects based on the series of relational object representations.

15. The at least one non-transitory computer readable storage medium of claim 14, wherein the instructions, when executed, further cause the computing system to:

include real-time perceptual error information with the predicted object trajectories; and

modify behavior of an autonomous vehicle based on the predicted object trajectories and the real-time perceptual error information.

**16**. The at least one non-transitory computer readable storage medium of claim **14**, wherein the instructions, when executed, further cause the computing system to:

determine deviation of observed object behaviors from predicted object behaviors; and

modify behavior of an autonomous vehicle based on the determined object behavioral deviation.

**17**. The at least one non-transitory computer readable storage medium of claim **14**, wherein the object trajectory histories include coordinates for a plurality of vehicles within a time window, wherein the series of time-stamped object graphs assist learning how the vehicles relate over the time window, wherein the relational object representations represent learned relationships among the plurality of vehicles over the time window, and wherein the first neural network is to encode location-based driving norms.

**18**. The at least one non-transitory computer readable storage medium of claim **17**, wherein the second neural network comprises a first recurrent neural network that is to encode temporal vehicle location changes and a second recurrent neural network that is to predict future behaviors for the plurality of vehicles.

**19**. The at least one non-transitory computer readable storage medium of claim **18**, wherein the first neural network comprises a graph attention (GAT) network and the second neural network comprises a long short-term memory (LSTM) network, and wherein the first neural network and the second neural network are trained as a unit using object trajectory histories generated from relational object data obtained from vehicle driving data collected across a plurality of geographic locations.

**20**. A relational reasoning method comprising:

generating a series of time-stamped object graphs based on object trajectory histories derived from external object data for a plurality of external objects;

generating, via a first neural network, a series of relational object representations based on the series of time-stamped object graphs; and

determining, via a second neural network, a prediction of future object trajectories for the plurality of external objects based on the series of relational object representations.

**21**. The method of claim **20**, further comprising:

including real-time perceptual error information with the predicted object trajectories; and

modifying behavior of an autonomous vehicle based on the predicted object trajectories and the real-time perceptual error information.

**22**. The method of claim **20**, further comprising:

determining deviation of observed object behaviors from predicted object behaviors; and

modifying behavior of an autonomous vehicle based on the determined object behavioral deviation.

**23**. The method of claim **20**, wherein the object trajectory histories include coordinates for a plurality of vehicles within a time window, wherein the series of time-stamped object graphs assist learning how the vehicles relate over the time window, wherein the relational object representations represent learned relationships among the plurality of vehicles over the time window, and wherein the first neural network encodes location-based driving norms.

**24**. The method of claim **23**, wherein the second neural network comprises a first recurrent neural network that encodes temporal vehicle location changes and a second recurrent neural network that predicts future behaviors for the plurality of vehicles.

**25**. The method of claim **24**, wherein the first neural network comprises a graph attention (GAT) network and the second neural network comprises a long short-term memory (LSTM) network, and wherein the first neural network and the second neural network are trained as a unit using object trajectory histories generated from relational object data obtained from vehicle driving data collected across a plurality of geographic locations.

* * * * *